

Obesity-Induced Microbiome Alterations Result in Severe Gastrointestinal Graft-Versus-Host Disease Following Allogeneic Hematopoietic Stem Cell Transplantation

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Allogeneic hematopoietic stem cell transplantation (allo-HSCT) remains a viable treatment option for many cancers but its clinical utility is limited due to the occurrence of graft-versus-host disease (GVHD). Understanding the impact of obesity on immune function has become increasingly important in the

setting of the current obesity pandemic. We report here that obesity has a negative and selective impact on acute gut GVHD. Diet-induced obese (DIO) mice exhibited increased gut permeability, endotoxin translocation and radiation-induced gastrointestinal damage. After allo-HSCT, DIO recipients across strains and sex had markedly increased pro-inflammatory cytokines (IL-6, TNF), GVHD biomarker ST2, MHC class II expression and exhibited rapid mortality associated with severe acute gut pathology. This obesity-associated lethal acute gut GVHD was dependent on donor CD4 T cells and occurred even in minor MHC mismatch strain combination in which only a delayed skin chronic GVHD resulted in lean recipients. Pro-inflammatory cytokine blockade targeting both IL-6 and TNF ameliorated obesity-associated acute gut GVHD while maintaining graft-versus-tumor (GVT) effects. Microbiome assessment of DIO mice revealed markedly reduced microbiome diversity and decreased *Clostridiaceae* abundance. Additionally, DIO mice had a significant increase of GVHD-associated *Akkermansia muciniphila* before and after allo-HSCT compared to the controls. Extended antibiotic treatment of DIO mice protected from the endotoxin translocation, cytokine storm as well as gut GVHD pathology but did not protect later development of chronic skin GVHD. These results demonstrate that obesity alters the microbiome and imparts differential effects on GVHD following allo-HSCT with decreased survival and this inferior outcome can be pre-empted by combined pro-inflammatory cytokine blockade or antibiotic pretreatment.

Disclosures

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Author notes

*Asterisk with author names denotes non-ASH members.

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